



**NANYANG  
TECHNOLOGICAL  
UNIVERSITY**  
SINGAPORE

## Lecture 4: CA6121 AI in Project Management

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# Project Cost Management, Optimization, Prediction, Forecasting

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# OBJECTIVES

- Overview of Project Cost Management
- Prediction and forecasting
- Earned Value Management (EVM)
- Cost baseline & project funding
- AI tools in project management (cost)
- Challenges / Benefits of using AI in cost estimation
- Case Studies



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# Project Cost Management

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Project Cost Management



# Cost Management Processes

- 1. Plan Cost Management**
- 2. Estimate Costs**
- 3. Determine Budget**
- 4. Control Costs**



# Cost Management Plan: Key Inputs

## Inputs

### 1. Project Management Plan

- Schedule management plan
- Risk management plan

### 2. Project Charter

### 3. Enterprise Environmental Factors

- Market Conditions (Products, Source and availability).
- Currency exchange rates.
- Published data.

### 4. Organizational process assets

- Internal procedures.
- Reporting, databases, accounting codes etc.



# Cost Management Plan: **Components**

## Outputs

- Cost Management Approach & Processes
- Roles and Responsibilities for Cost Management Activities
- Reporting Formats
- Units of measure
- Level of precision (decimal round up 100.49 to 100)
- Level of accuracy (+/- 10% or +/- 25% ??)
- Procedures to be followed
- Controls threshold
- Performance Management (Earned Value Management – EVM)
- Additional Details (Strategic funding, currency fluctuations)



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# Estimate Cost

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Project Cost Management





# Preparing Project Cost Estimates

- Cost Estimates are derived from Cost of Resources and includes:
  - Materials
  - Equipment
  - Labor
  - Services
  - Facilities

| Item | Type of Estimate         | Accuracy Range    | Remarks                                 |
|------|--------------------------|-------------------|-----------------------------------------|
| 1    | Rough order of Magnitude | +/- 25% or higher | Early-stage estimate                    |
| 2    | Definitive Estimate      | +/- 5% to +/-10%  | More clarity on resources by this point |



# Inputs to Cost Estimate

## Inputs

- Project Scope Statement
- Work Breakdown Structure (WBS) information on deliverables and work packages
- Project Schedule
- Project Resources
- Market Conditions
- Published commercial information and Cost Databases
- Organization templates, Historical Information, Lessons
- Risk Register (information on uncertainties and un-known)



# Project Cost Estimating Processes

## Tools & Techniques

- Project Team is involved
- Project requirements are clearly understood
- Cost Estimate can be developed using:
  - Manually
  - Software

Cost Estimating  
Method

**Top-Down**  
Estimating  
Method

**Bottom-up**  
Estimating  
Method



# Top-Down Estimating Method

**Top-Down**  
Estimating  
Method

## Tools & Techniques

- Relies on cost information from **past projects**
- Normally used at the **conceptual phase**
- Helps in making strategic decisions when **detailed information is not available**
- Method is **Quick and Less expensive**
- Estimate **level of Accuracy is Low**



## Bottom-up Estimating Method

### Bottom-up

Estimating  
Method

#### Tools & Techniques

- Builds up the cost from the cost of the lowest component
- Method is used when detailed information is available
  - WBS (Work Breakdown Structure) is done
  - Work Packages are known
  - Resource requirements can be established
- Method is **time consuming and more expensive**
- Estimate **level of Accuracy is High**



# Bottom-up Estimating Method

**Bottom-up**

Estimating  
Method

Tools &  
Techniques

Following are possible ways to develop Bottom-up Estimates:

- Direct Approach
- Parametric Estimating or Ratio Method
- Three-Point Estimating



# Bottom-up Estimating Method

**Bottom-up**

Estimating  
Method

Tools &  
Techniques

## Direct Approach

Derive Resource Cost or Work item from the cost components:

- Labour
- Materials
- Equipment
- Software / Tools
- Overheads
- Profits
- Contingency



# Bottom-up Estimating Method

**Bottom-up**

Estimating  
Method

Tools &  
Techniques

## Parametric Estimating or Ratio Method

This method uses statistical ration derived from historical information over time to get unit cost of:

- Work items
  - Materials
  - Services
- 
- Example:
    - Cost per AI Agent development for emails or notes of meeting compilation.
    - Cost per cubic metre of concrete.





# Bottom-up Estimating Method

## Bottom-up

Estimating  
Method

### Tools & Techniques

## Three-Point Estimating

- This method also known as **PERT** Method (**Program Evaluation and Review Technique**).
- Used when there is some degree of uncertainty about the work.
- Uses following three types of estimates:
  - **Most Likely Estimate**, which is based on the **realistic-case scenario** for the cost of the work item.
  - **Optimistic Estimate**, which is based on the **best-case scenario** for the cost of the work item.
  - **Pessimistic Estimate**, which is based on the **worst-case scenario** for the cost of the work item.



# Bottom-up Estimating Method

## Bottom-up

Estimating  
Method

### Tools & Techniques

## Three-Point Estimating

- **Triangular Distribution:**

$$cE = (cO + cM + cP) / 3$$

- **Beta Distribution Formula below.**

$$cE = (cO + 4cM + cP)/6$$

*cE is the Expected Cost Estimate,*

*cO is the Optimistic Cost Estimate,*

*cM is the Most Likely Cost Estimate*

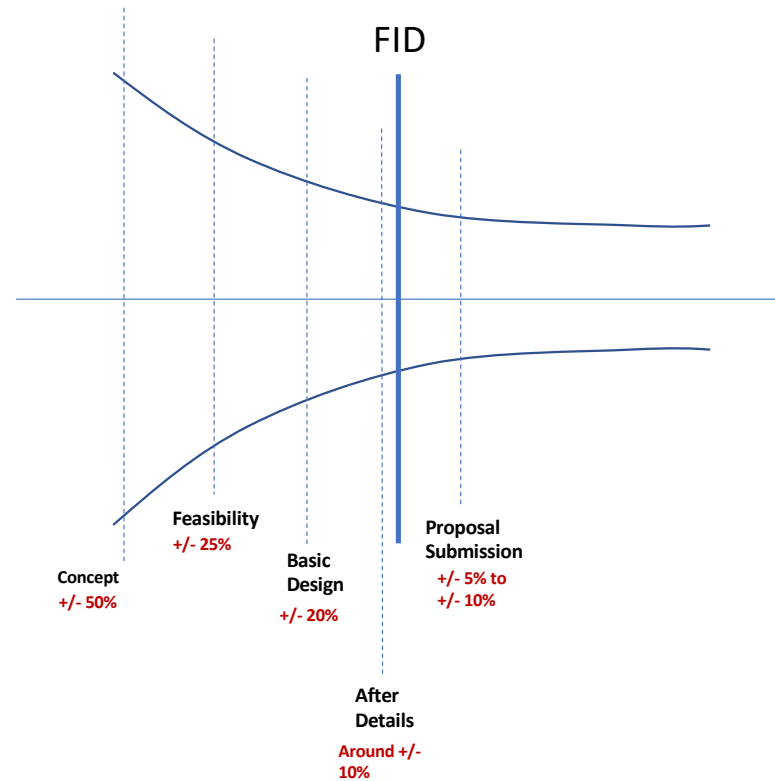
*cP is the Pessimistic Cost Estimate.*



# Estimate Accuracy

Outputs

Example



FID: Final Investment Decision



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# Determine Budget

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Project Cost Management



# Determine Budget

## Inputs

### 1. Project Management Plan

- Cost Management Plan
- Resource Management Plan
- Scope Baseline

### 2. Project Documents

- Basis of estimates
- Cost estimates
- Project Schedule
- Risk Register

### 3. Business documents

- Business case (critical business factors including financial success factors)
- Benefits management plan (NPV calculations, timeframe)

### 4. Agreements

### 5. Organizational process assets



## Determine Budget: Tools and Techniques

### Tools & Techniques

- a. Expert Judgement and Experience
- b. Cost Aggregation
- c. Data Analysis
  - Reserve Analysis
- d. Historical information review
- e. Funding Limit Reconciliation
- f. Financing



# Project Budget Template

**Source:**  
**Introduction to Project Management**  
 A Source Book for Traditional PM Basics  
**Davies A. Igberaese**

Prepared by: Ravish Maqsood  
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| PROJECT BUDGET                      |                             |          |               |                |               |                        |                    |                         |               |                  |
|-------------------------------------|-----------------------------|----------|---------------|----------------|---------------|------------------------|--------------------|-------------------------|---------------|------------------|
| Summary Cost of the project         |                             |          |               | Amount (in \$) |               | Details of the Project |                    |                         |               |                  |
| Total Budgeted Cost during the Peri |                             |          |               | -              |               | Name of the Company    |                    |                         |               | -                |
| Total Actual Cost during the period |                             |          |               | -              |               | Project Name or ID     |                    |                         |               | -                |
| Total Variance during the period    |                             |          |               | -              |               | Project Manager        |                    |                         |               | -                |
|                                     |                             |          |               |                |               | Start Date             |                    |                         |               | -                |
| S. No                               | Particulars                 | Material |               | Labour         |               | Fixed Cost             | Miscellaneous Cost | Budgeted Amount (in \$) | Actual Amount | Variance (in \$) |
|                                     |                             | Unit     | Cost per Unit | Hours          | Cost per Hour |                        |                    |                         |               |                  |
|                                     | Deliverable A               |          |               |                |               |                        |                    |                         |               |                  |
| 1                                   | Work Package 1              |          |               |                |               |                        |                    |                         |               |                  |
| 2                                   | Work Package 2              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| 3                                   | Work Package 3              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| 4                                   | Work Package 4              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| 5                                   | Work Package 5              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| (A)                                 | Sub Total                   |          |               |                |               |                        |                    |                         |               |                  |
|                                     | Deliverable B               |          |               |                |               |                        |                    |                         |               |                  |
| 1                                   | Work Package 1              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| 2                                   | Work Package 2              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| 3                                   | Work Package 3              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| (B)                                 | Sub Total                   |          |               |                |               |                        |                    |                         |               |                  |
|                                     | Deliverable C               |          |               |                |               |                        |                    |                         |               |                  |
| 1                                   | Work Package 1              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| 2                                   | Work Package 2              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| 3                                   | Work Package 3              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| 4                                   | Work Package 4              | -        | -             | -              | -             | -                      | -                  | -                       | -             | -                |
| (C)                                 | Sub Total                   |          |               |                |               |                        |                    |                         |               |                  |
| (D)                                 | Total of Project (A + B+ C) |          |               |                |               | -                      | -                  | -                       | -             | -                |



# Determine Budget: Tools and Techniques

## Outputs

- a. Cost Baseline
- b. Project Funding Requirements
- c. Project Documents Updates
  - Cost Estimates
  - Project Schedule
  - Risk Register





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# Control Cost

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Project Cost Management



## Control Costs

It's a process of

**Monitoring** the status of the project **to update the project costs**  
**and managing changes** to the cost baseline.



# Control Costs

## Inputs

### 1. Project Management Plan

- Cost Baseline
- Cost Management Plan
- Performance Measurement Baseline

### 2. Project Documents

- Lessons register

### 3. Project Funding Requirements

### 4. Work Performance Data

- Project Progress
- Cost authorized and incurred

### 5. Organizational process assets

- Procedures updates
- Lessons
- Tools



# Control Costs: Tools and Techniques

## Tools & Techniques

### a. Expert Judgement

- Variance analysis
- Earned value analysis
- Forecasting
- Financial analysis

### b. Data Analysis

### c. To-Complete Performance Index (TCPI)

### d. Performance Management Information System (PMIS)



# Control Costs: Tools and Techniques

## b. Data Analysis

- **Earned Value Management (EVM)**
  - Planned Value
  - Earned Value
  - Actual Cost
- **Variation Analysis**
  - Schedule Variance
  - Cost Variance
  - Schedule performance index
  - Cost performance index
- **Trend Analysis**
  - Charts
  - Forecasting
- **Reserve Analysis**



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# Cost prediction, optimization, and forecasting

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Project Cost Management



## Earned Value Management (EVM)

Earned Value Management (EVM) is a **methodology that combines scope, schedule, and resource measurements to assess project performance and progress.**

**It integrates scope baseline and cost baseline, along with the schedule baseline to form the performance management.**



## Earned Value Management (EVM)

Earned Value Management (EVM) develops and monitor 3 key dimensions for each work package and control account:

- 1. Planned Value (PV)**
- 2. Earned Value (EV)**
- 3. Actual Cost (AC)**





## Earned Value Management (EVM)

Variances from approved baseline are also monitored:

- 1. Schedule Variance (SV)**
- 2. Cost Variance (CV)**
- 3. Schedule Performance Index (SPI)**
- 4. Cost Performance Index (CPI)**



# Determine Budget: Tools and Techniques

## Outputs

### a. Work Performance Information

- Calculated CV, SV, CPI, SPI, TCPI, VAC values for WBS

### b. Cost Forecasts

- Estimates At Completion (EAC) or a Bottom-up EAC documented and communicated

### c. Change Requests

### d. PMP Updates

- Cost Baseline
- Cost Management Plan
- Performance management baseline

### e. Project Documents Updates

- Assumption log
- Basis of estimates
- Cost estimates
- Risk register



## Earned Value Management (EVM) Example

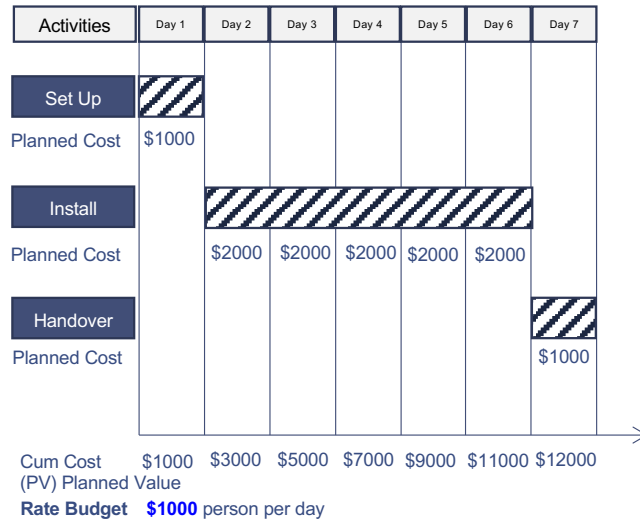
A company plan to install new computers supply and install by the supplier as follows:

| Day | Work           | Person<br>required (no.) | Planned Rate<br>(\$) | Planned Cost<br>(\$) |
|-----|----------------|--------------------------|----------------------|----------------------|
| 1   | Setup          | 1                        | \$1000               | \$1000               |
| 2   | Installation   | 2                        | \$1000               | \$2000               |
| 3   | Installation   | 2                        | \$1000               | \$2000               |
| 4   | Installation   | 2                        | \$1000               | \$2000               |
| 5   | Installation   | 2                        | \$1000               | \$2000               |
| 6   | Installation   | 2                        | \$1000               | \$2000               |
| 7   | Final handover | 1                        | \$1000               | \$1000               |

How do we set this information to monitor?



## Earned Value Management (EVM) Example



**This was the Plan**  
**But what happened in actual?**



## Earned Value Management (EVM) Example

ACTUAL Scenario: Company **could not find person at planned rates:**

Work done **1<sup>st</sup> day is 100%** and **2<sup>nd</sup> day 50% by 2 people.**

| Day | Work           | Person<br>required<br>(no.) | Planned Rate<br>(\$) | Planned<br>Cost (\$) | Actual<br>Rate (\$) | Actual<br>Cost (\$) |
|-----|----------------|-----------------------------|----------------------|----------------------|---------------------|---------------------|
| 1   | Installation   | 1                           | \$1000               | \$1000               | \$1100              | \$1100              |
| 2   | Installation   | 2                           | \$1000               | \$2000               | \$1500              | \$3000              |
| 3   | Installation   | 2                           | \$1000               | \$2000               |                     |                     |
| 4   | Installation   | 2                           | \$1000               | \$2000               |                     |                     |
| 5   | Installation   | 2                           | \$1000               | \$2000               |                     |                     |
| 6   | Installation   | 2                           | \$1000               | \$2000               |                     |                     |
| 7   | Final handover | 1                           | \$1000               | \$1000               |                     |                     |

How do we represent this information?



## Earned Value Management (EVM) Example

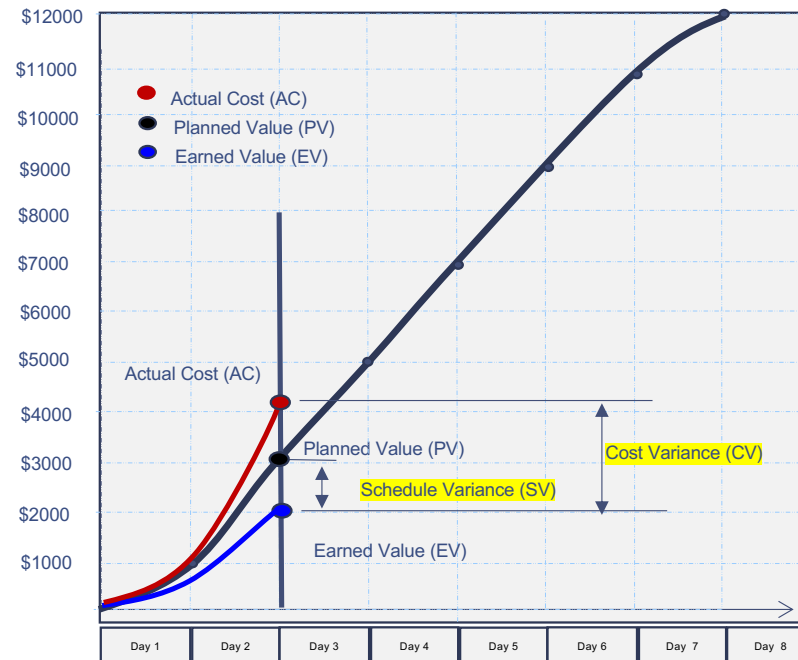
### **Calculate:**

- 1. Schedule Variance (SV)**
- 2. Cost Variance (CV)**
- 3. Schedule Performance Index (SPI)**
- 4. Cost Performance Index (CPI)**



# Earned Value Management (EVM) Example

| Activities     | Day 1                                                | Day 2  | Day 3  | Day 4  | Day 5  | Day 6   | Day 7   |
|----------------|------------------------------------------------------|--------|--------|--------|--------|---------|---------|
| Set Up         | ▨                                                    |        |        |        |        |         |         |
| Planned Cost   | \$1000                                               |        |        |        |        |         |         |
| Install        |                                                      | ▨      | ▨      | ▨      | ▨      | ▨       |         |
| Planned Cost   |                                                      | \$2000 | \$2000 | \$2000 | \$2000 | \$2000  |         |
| Handover       |                                                      |        |        |        |        |         | ▨       |
| Planned Cost   |                                                      |        |        |        |        |         | \$1000  |
| Cum Cost (PV)  | \$1000                                               | \$3000 | \$5000 | \$7000 | \$9000 | \$11000 | \$12000 |
| Rate Budget    | \$1000 person per day                                |        |        |        |        |         |         |
| Work Done      | 100%                                                 | 50%    |        |        |        |         |         |
| EV             | \$1000                                               | \$1000 |        |        |        |         |         |
| Cum (EV)       | \$1000                                               | \$2000 |        |        |        |         |         |
| Rate Actual    | \$1100 1 <sup>st</sup> day and \$1500 person per day |        |        |        |        |         |         |
| Actual Cost AC | \$1100                                               | \$3000 |        |        |        |         |         |
| Cum (AC)       | \$1100                                               | \$4100 |        |        |        |         |         |



$$\begin{aligned} \text{Cost Variance (CV)} &= \text{EV} - \text{AC} \Rightarrow \$2000 - \$4100 = -\$2100 \\ \text{Schedule Variance (SV)} &= \text{EV} - \text{PV} \Rightarrow \$2000 - \$3000 = -\$1000 \end{aligned}$$

+ve value is good as it means you have earned more than planned.  
-ve value is BAD as it means you are falling behind.

$$\begin{aligned} \text{Cost Performance Index (CPI)} &= \text{EV} / \text{AC} \Rightarrow \$2000 / \$4100 = 0.48 \\ \text{Schedule Performance Index (SPI)} &= \text{EV} / \text{PV} \Rightarrow \$2000 / \$3000 = 0.67 \end{aligned}$$

< 1 is not good, means you are behind



# Forecasting

Forecast is developed to:

**Estimate At Completion (EAC)** that may differ from **Budget At Completion (BAC)**.

EAC is typically based on:

Actual Costs (AC) incurred to date for the work completed

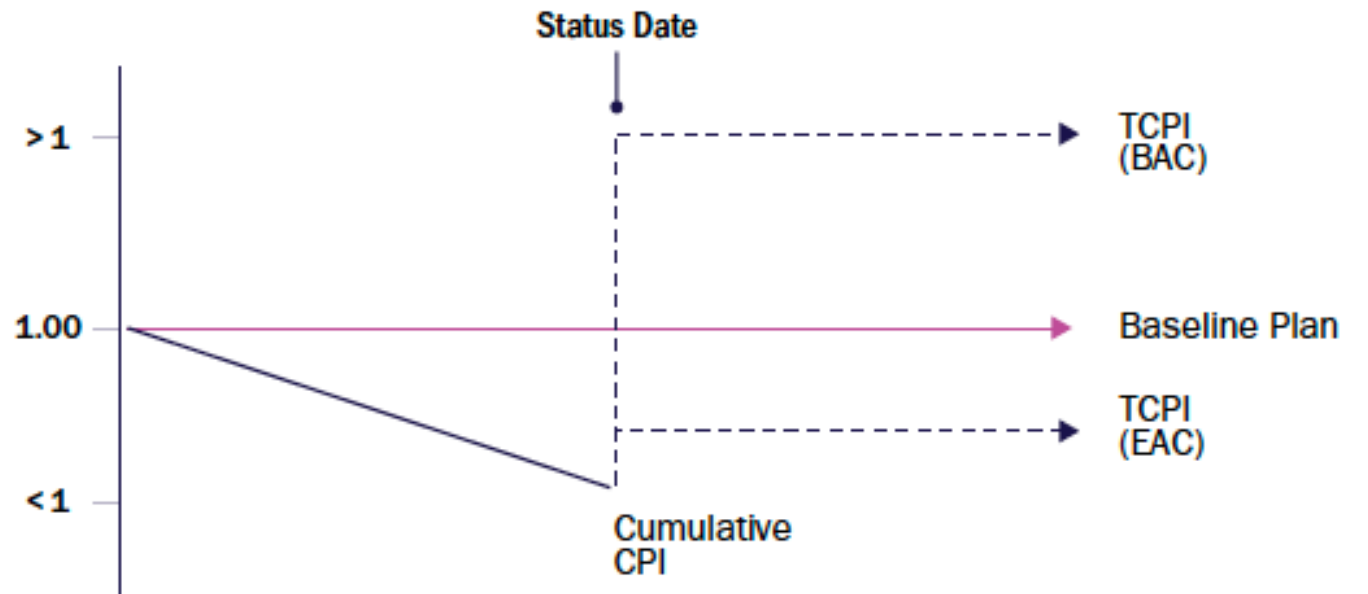
**+** (Plus)

Estimate To Complete (ETC)





# Forecasting



## Formula:

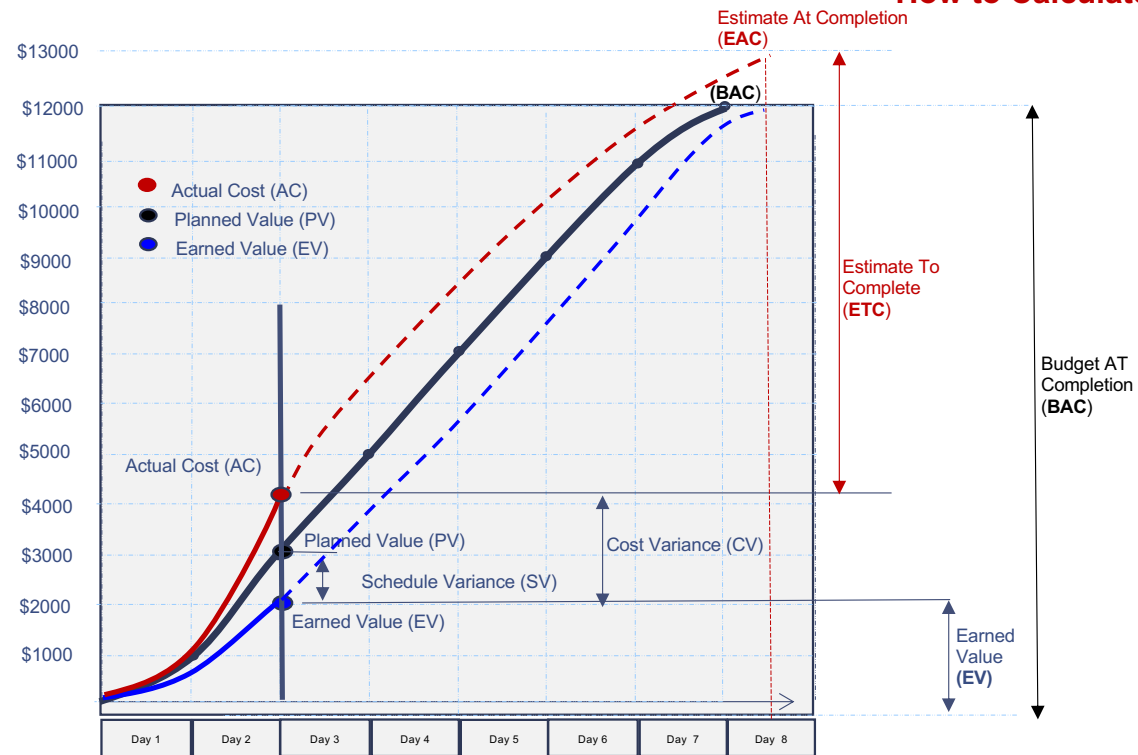
$$\frac{\text{Work Remaining (BAC - EV)}}{\text{Funds Remaining (BAC - AC) or (EAC - AC)}} = \text{TCPI}$$

Figure 5-24. To-Complete Performance Index (TCPI)



# Forecasting

## How to Calculate EAC?



Estimate AT Completion => **EAC**  
Budget AT Completion => **BAC**  
Estimate To Complete => **ETC**



# Forecasting

## 1. EAC forecast for ETC work performed at budgeted rate.

When the work is to be completed at Budgeted Rate.

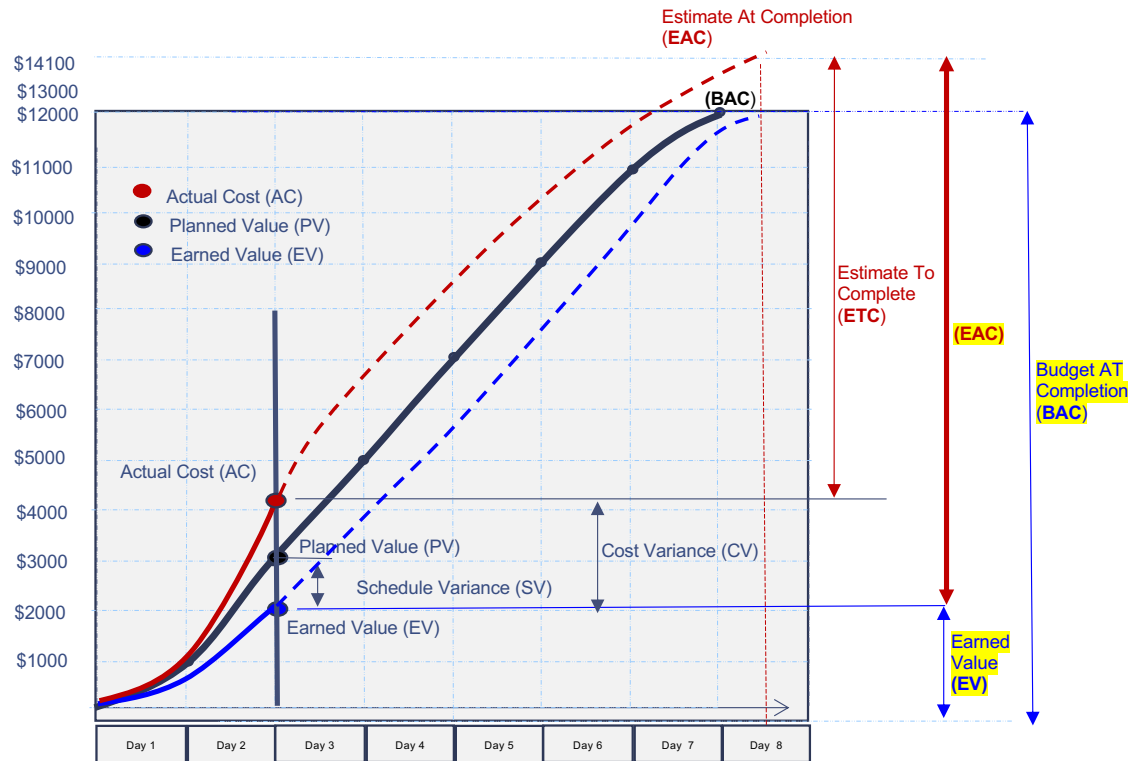
- whatever happened day 2 is only one time scenario and now onwards the project will fall back on budgeted rate.

$$\begin{aligned} \text{EAC} &= \text{AC} + (\text{BAC} - \text{EV}) && (\text{or } \text{AC} + \text{ETC}) \\ &= \$4100 + (\$12,000 - \$2000) \\ &= \$14,100 \end{aligned}$$

Means if we continue current scenario, we would be spending \$14,100 to complete the project.



# Forecasting



$$EAC = AC + (BAC - EV) \quad \text{or} \quad AC + ETC$$

$$= \$4100 + (\$12,000 - \$2000)$$

$$= \$14,100$$

Estimate AT Completion => **EAC**  
 Budget AT Completion => **BAC**  
 Estimate To Complete => **ETC**



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# Cost Baseline & Project Funding

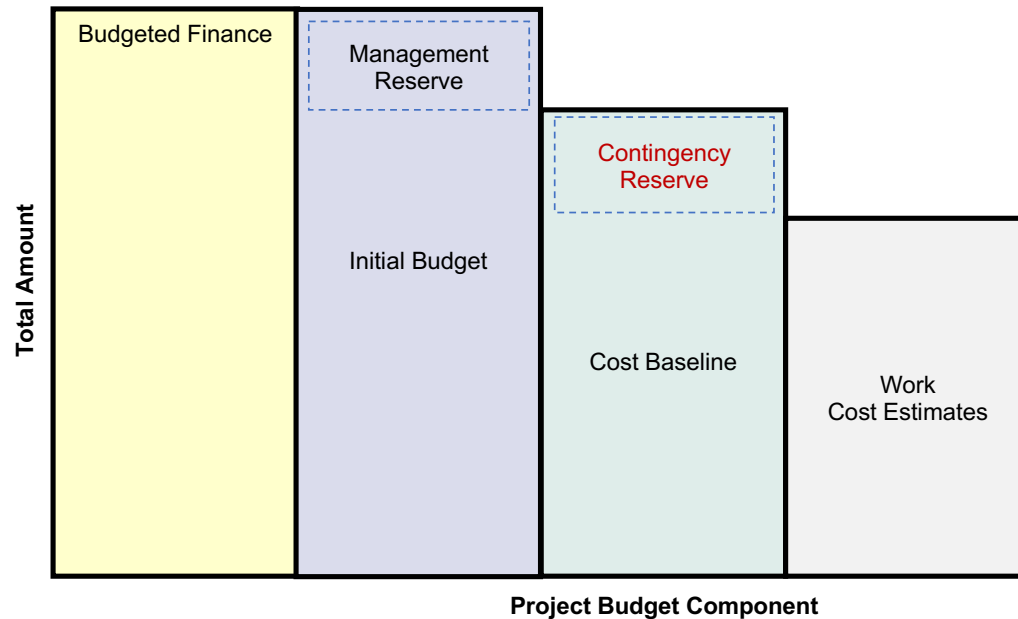
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Project Cost Management



# Budget Buildup: Initial Budget, Cost Baseline, Work Cost Estimate

*Reserves Managed Implicitly*

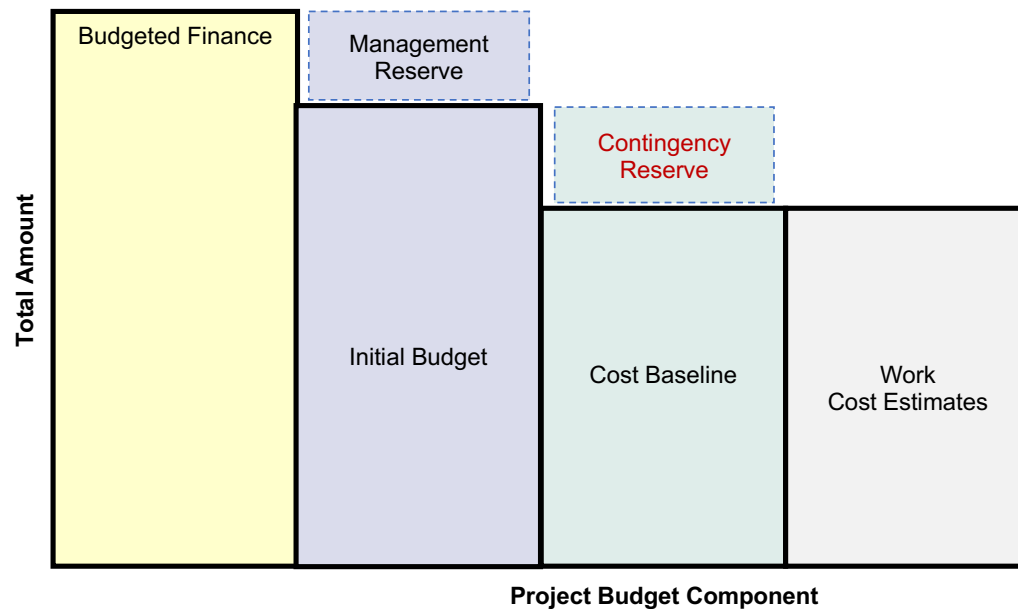


Source: PM®BOK 8<sup>th</sup> Edition



# Budget Buildup: Initial Budget, Cost Baseline, Work Cost Estimate

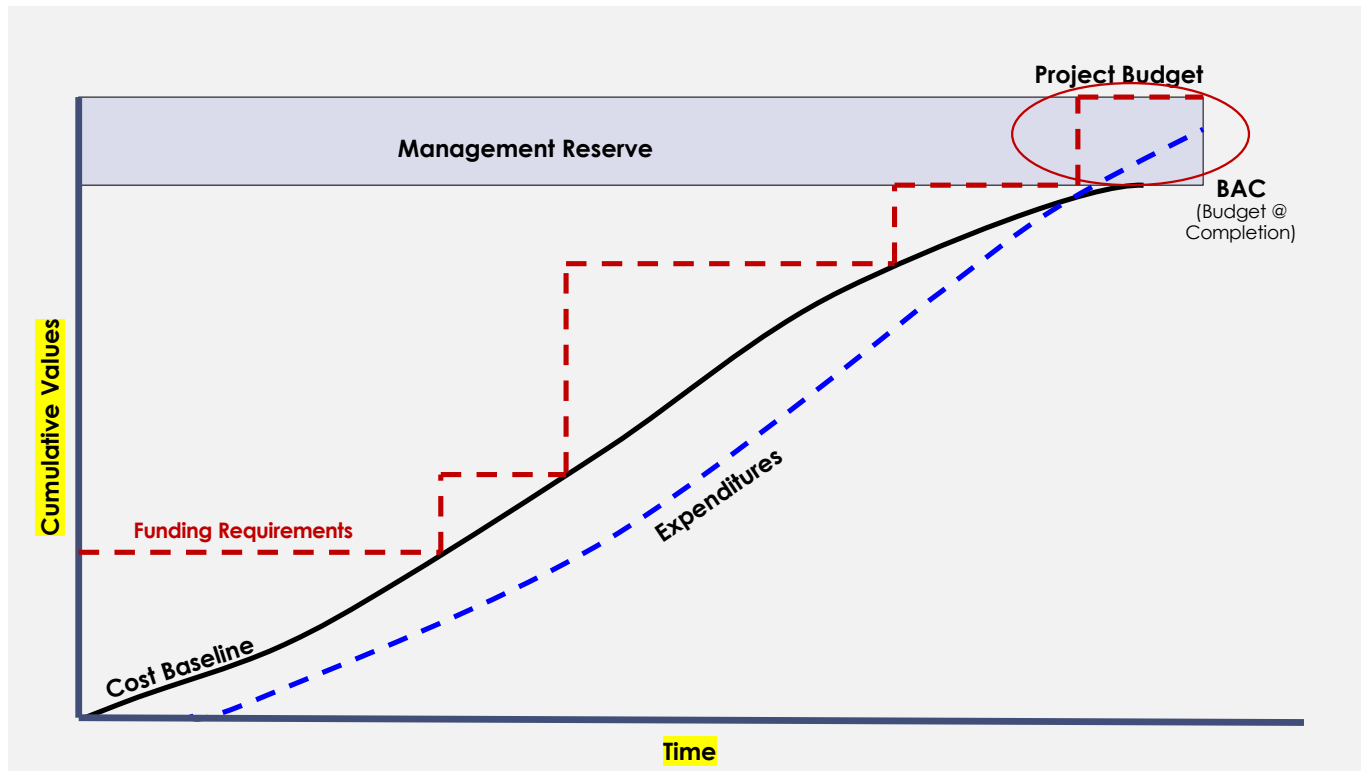
*Reserves Managed Explicitly*



Source: PM®BOK 8<sup>th</sup> Edition



# Cost Baseline



Source: PM®BOK 6<sup>th</sup> Edition





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# AI Tools in Project Management

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High level Perspective



## AI tools in Project Management

**AI Tool can help**

- Using AI to review the Cost Data.
- Review against past projects.
- Identify the Risk Profile on this project calling for RFP.
- Review the cost in compliance to the policies.



# Machine Learning in Project Management

*AI Tool can help*

## Estimating

- Analysis of past projects and create a new estimate for future projects.
- Challenge is to build the accurate data set



# Machine Learning in Project Management

**AI Tool can help**

## Cost Estimates:

Can build up the estimates using the data on:

- Resource availability
- Resource Dependencies
- Resource Durations
- Project Constraints
  
- Parallel working
- Fast tracking
- Additional resources

Number of alternative scenarios possible for the given information & priorities



# AI tools in Project Management

**AI Tool can help**

- Cost optimization.
- Review the scope and optimize etc.
- Baseline Cost formats and usage of available AI Tool.
- Any changes will have some impact on the project cost.
- Creating a workflow to review the impact.
- Change request model can be embedded with approval processes.



# AI tools in Project Management

AI Tool can help

## AI Cost Estimation Tools Comparison

| Tool Name            | Industry Focus             | Key Features                                                                                                                            | Accuracy Rate | Integration Level | Best For                    |
|----------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|-----------------------------|
| Cleopatra Enterprise | Construction & Engineering | <ul style="list-style-type: none"><li>Historical data analysis</li><li>BIM integration</li><li>Material cost tracking</li></ul>         | 95%           | High              | Large construction projects |
| OpenCost             | Cross-Industry             | <ul style="list-style-type: none"><li>Open-source flexibility</li><li>Custom algorithms</li><li>Community support</li></ul>             | 88%           | Very High         | Budget-conscious teams      |
| Procore              | Construction               | <ul style="list-style-type: none"><li>Real-time tracking</li><li>Mobile-first design</li><li>Automated alerts</li></ul>                 | 92%           | High              | Construction management     |
| AutoML & Azure       | IT & Software              | <ul style="list-style-type: none"><li>Code analysis</li><li>Sprint planning</li><li>Resource optimization</li></ul>                     | 90%           | Very High         | Software development        |
| IBM Watson & SAP     | Manufacturing & Logistics  | <ul style="list-style-type: none"><li>Supply chain optimization</li><li>Predictive maintenance</li><li>Global market analysis</li></ul> | 94%           | Moderate          | Enterprise operations       |



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# Challenges using AI in Cost Estimation

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Project Cost Management



## AI in Cost Estimation: Challenges

- Resistance to change.
- Availability of data due to confidentiality.
- Accuracy of the output which needs more human intervention.
- Lack of properly built and calibrated model.
- Insufficient records in training or testing datasets.
- With more complex AI models, more different datasets might be required.





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# Benefits of using AI in Cost Estimation

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Project Cost Management



# AI in Cost Estimation: Challenges

- Improved accuracy
- Improved efficiency.
- Enhanced decision-making
- Ability to adapt quickly to changing project conditions.



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# Prediction and forecasting

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Project Planning & Scheduling / Cost Management / Resource Management



## Prediction and forecasting

- Earned Value Management covers prediction and forecasting.
- Any changes may have an impact on schedule, cost, risk, and resources.



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# Case Studies

## AI in Project Management (Not in AI Project)

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AI in Project Management



## Case Study AA

**AI in Project  
Management**

### Case Study: Development of Project Cost Plan

**Scenario:** GenAI is used to generate the business case documentation.

**Tools used:** LLM via ChatGPT

#### Concerns:

- Confidential information data could not be used.
- Need human intervention to review the outputs.
- Output was not in the existing format available in the company.
- Project Manager still needed time to review the outcome.

#### Solutions:

- Inhouse LLM (ChatGPT similar) model will help to resolve confidentiality issues.
- Implement human-in-the-loop review for AI alerts
- Provide training to build trust among workers



## Case Study BB

**AI in Project  
Management**

### Case Study: Calculate EVM (Earned Value Management)

**Scenario:** GenAI is used to generate the business case documentation.

**Tools used:** LLM via ChatGPT

#### Concerns:

- Generation of Chart needed more interfaces.
- AI Agent would be of better help.
- Confidential information data could not be used.
- Need human intervention to review the outputs.
- Project Manager still needed time to review the outcome.

#### Solutions:

- Inhouse LLM (ChatGPT similar) model will help to resolve confidentiality issues.
- Provide training to build trust among workers



## Student Exercise 3

- Perform a search or collate your own personal experience on Cost Estimation or EVM. Please elaborate your understanding where AI can help in the whole process from Project Management perspective.
- Any industry.
- Any location.
- Any type of project (IT, Construction, Pharma, etc.)

Prepare your response and submit via  
Submission Link Provided Separately.





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## End of class

